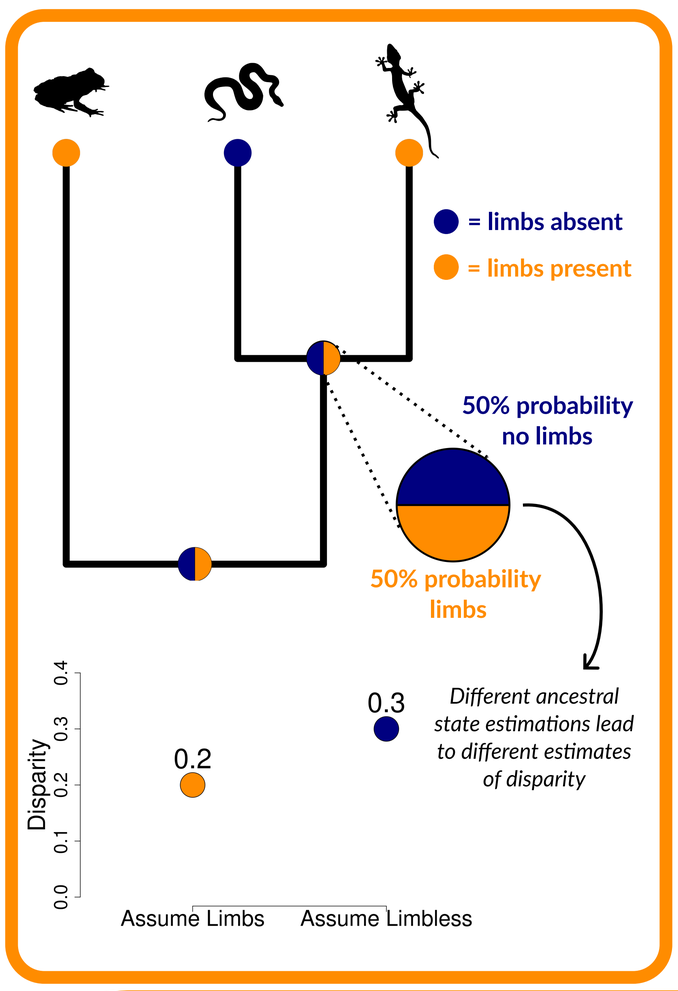
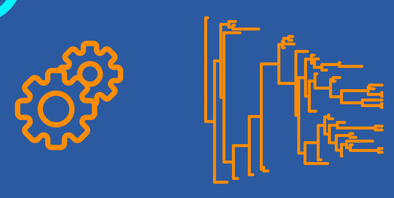
What is the effect of ancestral state estimation on disparity?

Caleb Scutt[1], Natalie Cooper[2], Gavin Thomas[1], Thomas Guillerme[1]
Contact: cnscutt1@sheffield.ac.uk
[1]University of Sheffield, [2]Natural History Museum, London.

University of Sheffield



1.1 Tree simulations



Simulated 100 trees using birth-death algorithm.

1.2 Mass extinction simulations

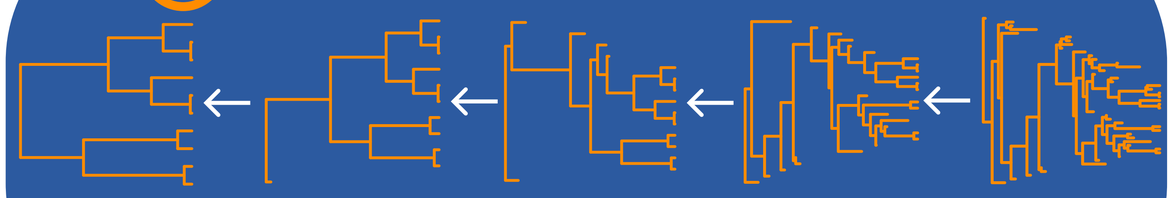
Mass extinction trees simulated with varying severity (75-95%).

2 Trait simulations

Fast | Medium | Slow

Discrete trait data simulated using three different transition rates across each tree.

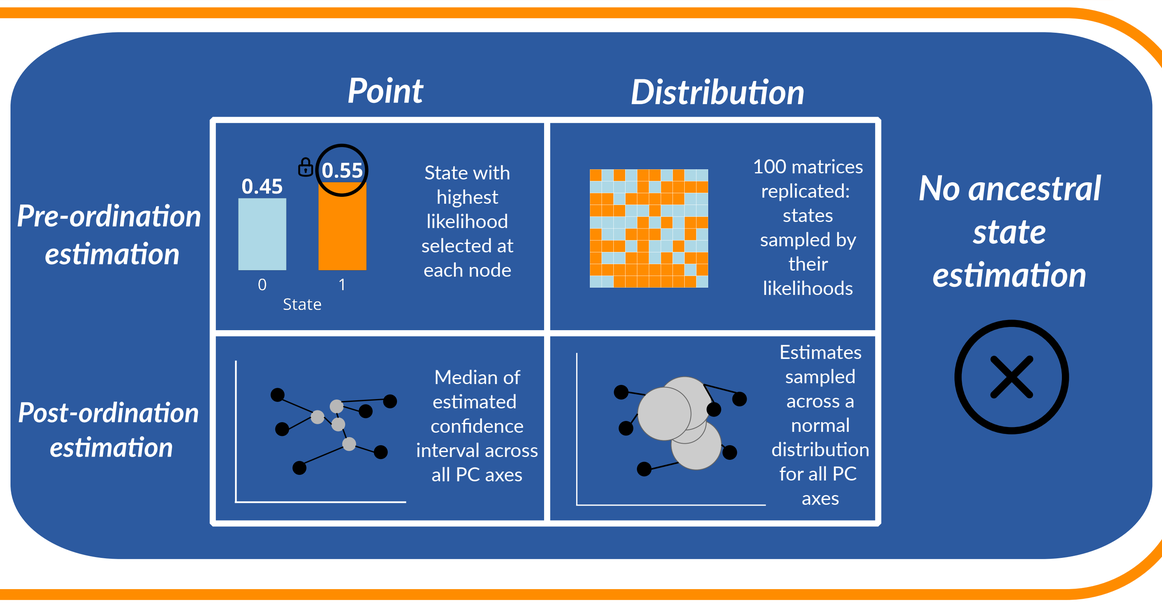
3 Fossil sampling



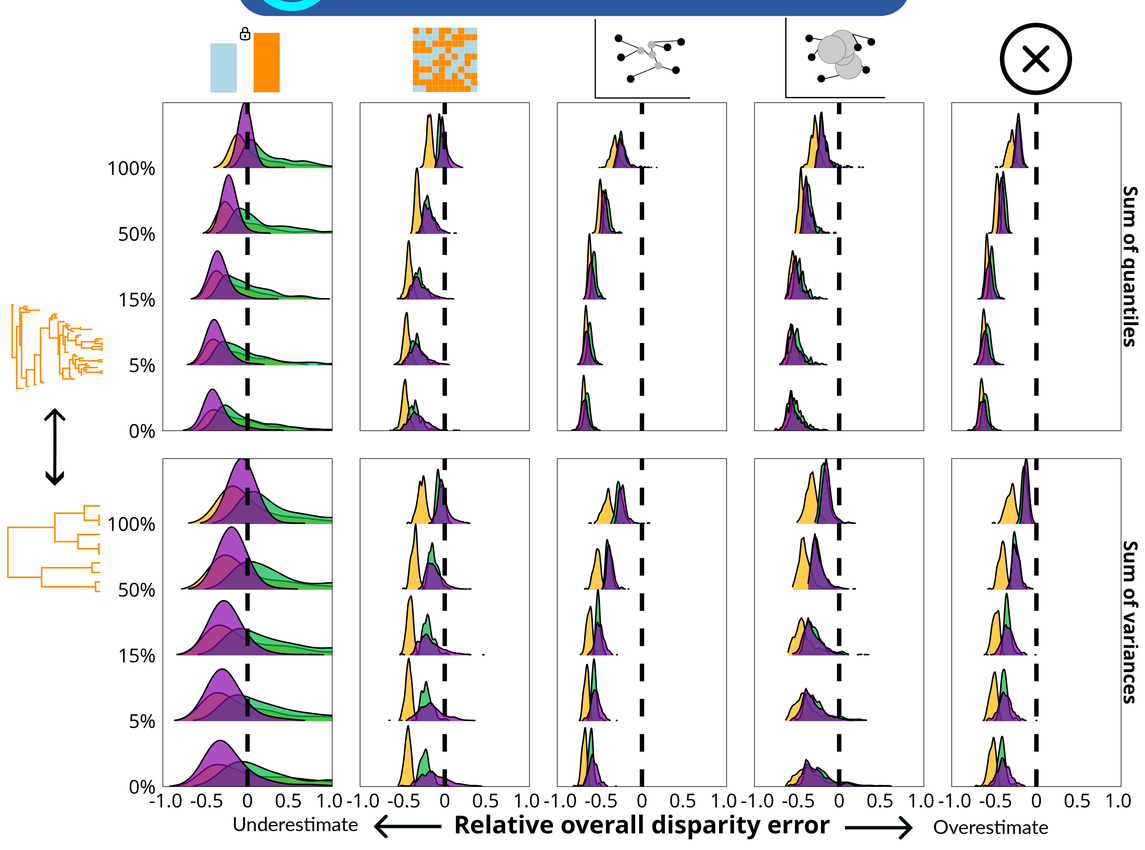
Prune each tree and matrix across a range of intensities to simulate different fossil sampling levels.

4 Ancestral state estimation

- Maximum Likelihood estimates the probability of each possible state at a node.
- A dominant state is simple to infer — but when states are equally likely, we face uncertainty.
- We used a combination of two different approaches for ancestral state estimation.
  - Pre- versus post-ordination estimation.
  - Point estimate versus distributions.
- This was compared against a null model of no ancestral state estimation, i.e., just using tip data.



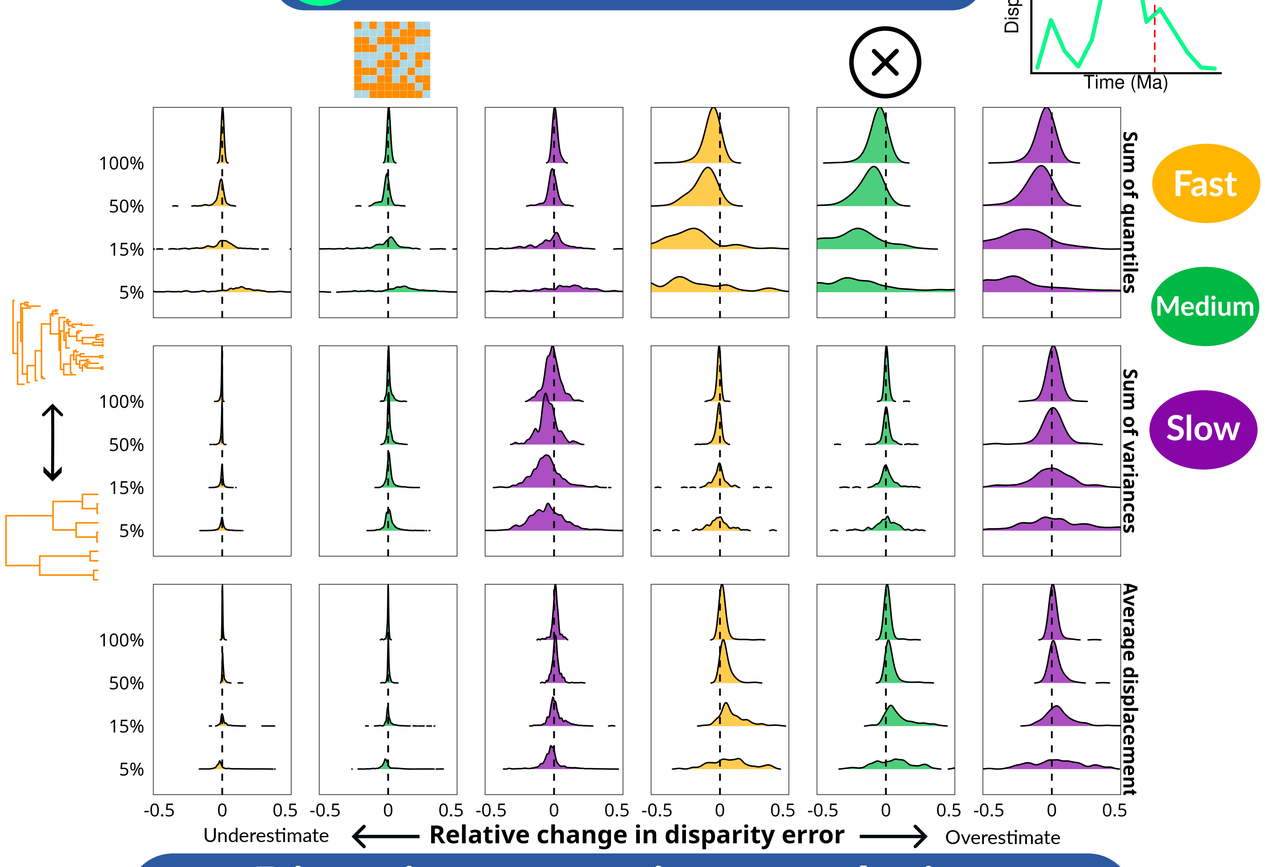
5.1 Overall disparity error



Overall disparity conclusions

- **Pre-ordination distribution method is the best** performing estimation method.
- **Increased fossil sampling improves recovery** of true disparity value.
- Omitting ancestral state estimations **always leads to underestimates of disparity.**

5.2 Disparity across time error



Disparity across time conclusions

- Similar pattern observed: **pre-ordination distribution method performs best** out of all estimation methods.
- **Increased fossil sampling has a more pronounced effect** for recovering accurate changes in disparity.
- More drastic changes in disparity **are harder to recover.**